Course Name: Digital Image Processing

Course Outcome

CO1: Understand mathematical formulation of an image, its processing steps and relationship between image pixels.

CO2: Apply Image enhancement using intensity transformations and spatial filtering.

CO3: Analyze image enhancement for frequency domain using Fourier transform.

CO4: Formulate region of interest through morphological operations.

CO5: Evaluate strongly co-related regions obtained through Segmentation using discontinuity and Homogeneity based segmentation techniques

CO6: Describe an object of an image using Shape Number and Boundary descriptors.

Printed Pages: 04

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End Term Examination, Odd Semester 2023-24

B.Tech. (CSE), Year-III, Semester-V

BCSE0101: Digital Image Processing

Time: 3 Hours

Maximum Marks: 50

Section – A

Attempt AllQuestions

4 X 5 = 20 Marks

<table>
<tr><th>No.</th><th>Detail of Question</th><th>Marks</th><th>CO</th><th>BL</th><th>KL</th></tr>
<tr><td>1</td><td>A common measure of transmission for digital data is the baud rate, defined as the number of bits transmitted per second. Transmission is accomplished in packets consisting of a start bit, a byte (8 bits) of information and a stop bit.
a) How many minutes would it take transmit a 1024x1024 image with 256 gray levels if we use a 56 k baud modem?
b) What would be the time required if we use a 850 k band transmission line?</td><td>4</td><td>1</td><td>U</td><td>C</td></tr>
<tr><td>2</td><td>Perform histogram stretching so that the new image has a dynamic range of [0, 7].
<table>
<tr><th>Gray level</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th></tr>
<tr><td>No. of pixels</td><td>0</td><td>0</td><td>50</td><td>60</td><td>20</td><td>10</td><td>40</td><td>0</td></tr>
</table></td><td>4</td><td>2</td><td>P</td><td>P</td></tr>
<tr><td>3</td><td>Suppose that a 3-bit image (L=8) of size 10 × 10 pixels (MN = 100) has the intensity distribution shown in the following table A. Get the histogram transformation function and make the output image with the specified histogram, listed in the table B.</td><td>4</td><td>2</td><td>P</td><td>C</td></tr>
</table>

Histogram A

Gray Level	0	1	2	3	4	5	6	7
No. of Pixels	4	17	15	18	24	12	0	10

Histogram B

Gray Level	0	1	2	3	4	5	6	7
No. of Pixels	0	0	0	36	24	12	8	20

4	Suppose two discrete one dimensional functions are represented by the sequences f = [5 7 11 8 2 6 8 9 7 4 3], h = [1 2 1]. Compute opening and closing.	4	4	P	P
5	There is a segmented image A in which an object has been detected. The detected object has boundary pixels, and inner pixels are missing which create a gap. To fill this gap, write an algorithm and apply to fill the inner region of the image A.	4	4	A	F

A

1	1	1	1	
1			1	
1	1		1	
	1		1	
	1	1	1	

Section – B

Attempt All Questions

3 X 5 = 15 Marks

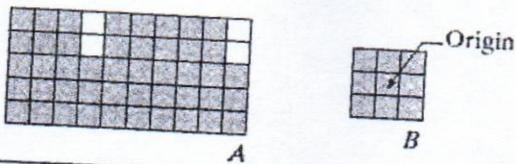
No.	Detail of Question	Marks	CO	BL	KL
6	Dilation & Erosion are duals of each other with respect to set complementation & reflection. Prove $(A \ominus B)^c = A^c \oplus \hat{B}$	3	4	U	C
7	There is an image A and structuring element B is given. Apply morphological operation erosion and dilation on image A by structuring element B.	3	4	A	C

1			
1			
	1	1	1
	1		
	1		

A

1	1

B

8	There is an image A and structuring element B. Write and apply an algorithm over the given image and structuring element to extract the boundary of the image A. (Gray color represents 1 and white color represents to 0) 	3	5	A	P
9	What are the different thresholding methods for segmentation? Write down basic global thresholding algorithm.	3	5	R	C
10	Image segmentation is the process of partitioning an image into multiple segments, or regions. The goal of image segmentation is to simplify the representation of an image into regions that is more meaningful and easier to analyze. How region growing works for segmentation? Explain with an example.	3	5	U	F

Section – C

*Attempt **All** Questions*

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	How region splitting and merge operation works for image segmentation? Explain with suitable example.	5	5	R	F
12	Find the following shape-description of the given shape (black circle is the origin): i. Chain code ii. First difference iii. Circular first difference iv. Shape number v. Order 	5	6	A	P
13	The purpose of edge detection in general is to significantly reduce the amount of data in an image, while preserving the structural properties to be used for	5	6	P	C

	further image processing. How the Canny Edge Detection reduce the amount of data to preserve structural properties.		